Soil Moisture Prediction Using Machine Learning Techniques

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ABSTRACT

Although - Soil moisture is the main factor in agricultural production and hydrological cycles, and its prediction is essential for rational use and management of water resources. However, soil moisture involves complicated structural characters and meteorological factors, and is difficult to establish an ideal mathematical model for soil moisture prediction. Prediction of soil moisture in advance will be useful to the farmers in the field of agriculture. In this paper, we have used machine learning techniques such as linear regression, support vector machine regression, PCA, and Naïve Bayes for prediction of soil moisture for a span of 12 to 13 weeks ahead. These techniques have been applied on four different datasets collected from 13 different districts of West Bengal, and four different crops (Potato, Mustard, Paddy, Cauliflower) collected over the span of about 1st January 2020 – 30th March 2020. The performance of the predictor is to be evaluated on the basis of F1-Score.

CCS CONCEPTS

• **Computing methodologies**; • **Machine learning**; • **Machine learning algorithms**; • **Feature selection**;

KEYWORDS

Stock Market, Machine Learning, Support Vector Machine, PCA, Linear Regression, Naïve Bayes, F1-score

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1 INTRODUCTION

India is a country where the majority of the population is dependent on agriculture for their livelihood. Indian soils are less fertile, especially in case of micronutrients. In recent years, they have seen that soil health is somehow related with the sustainability in the field of agriculture and also the current crop yield levels can be improved by maintaining the fertility of the soil. Water is the primary resource which determines the survival and development of the Earth's occupant. Soil moisture does not only plays an important role in maintenance of plant growth but also as a key link to the water cycle of soil-plant-atmosphere continuous systems. Accessibility

Following the guidelines throughout this template will also improve the accessibility of your manuscript and increase the audience for your work. Ensure that heading styles are applied as instructed, tables are created using Word's table feature (rather than an image), figures have a text equivalent, and list styles are applied as instructed.

1.1 Importance of soil moisture

The amount of moisture or soil water is important to be known because:

- Soil water serves as a solvent and food nutrients carrier to boost in plant growth
- The yield of a crop is determined by availability of the amount of water
- Soil water is a nutrient factor on its own
- Temperature of the soil is regulated by the movement of water in the soil
- On soil water, soil forming processes and weathering depends
- Micro-organisms metabolic activities require soil water
- Chemical and biological activities of soil are helped by soil water
- Soil is the principal constituent for the growth of plant
- Water is essential for photosynthesis.

1.1.1 Problem Statement. Soil moisture and its availability is to support plant growth is a primary factor in farm productivity, and too little moisture can lead to loss in plant death. Causing root disease and wasted water.

1.1.2 Objective.

1. To predict the soil moisture requirement for the upcoming next three months of a year by using the Machine Learning Algorithms.
2. To study the machine learning approach for detection of the patterns by using both supervised and unsupervised learning algorithms i.e. Naïve Bayes, SVM, Linear Regression and Principal Component Analysis.

The focus of this work is to improve the process of predicting the soil moisture through a machine learning approach. This raw data collected from the fields across the 13 districts of West Bengal has been used in this work to machine learning models to learn. It is found that machine learning models can be used to predict the soil moisture requirement in the upcoming years with significant accuracy depending on the temperature and humidity content of the atmosphere. The accuracies vary from one machine-learning algorithm to others. The accuracies saw a rise when the data was converted into time series supervised learning format, although

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with the application of the same algorithm. There is an average increase in the accuracy when the trend is predicted using time series supervised learning data. In the future, this work can be expanded by adding more raw data on various other crops depending on the quality of the soil across the country and calculations can be more technically indicating as features. For the time series implementation, this work uses the lag of the past 90 days, and this can be increased in the future. During experimentation, it was observed that the accuracies of classifiers varies for different crops; hence the work can also be carried out to find the best suitable classifier for each individual stock.

For future research, the predictive system should be extended to include rainfall forecasts. This will ensure that irrigation is optimized to further increase water savings through the maximum utilization of forecasted rainfall depths. The development of crop-specific models trained on a rich dataset obtained from sites with similar soil types will enhance the adoption of data-driven soil moisture models for use in irrigation scheduling applications.

One of the main goals of developing all these models is to produce reliably good estimates of Soil Moisture Content using the most easily available attributes such as the meteorological factors and vegetation indices. This information is very helpful to farmers and water researchers to understand the approximate conditions of the fields and guides them to decide more efficiently about farming practices such as irrigation and application of fertilizers.

1.2 Regarding the methods of machine learning

In machine learning, the computers are made to learn without being explicitly programmed. The code is not written for every specific problem; instead, data is provided to a machine learning algorithm, and subsequently, logic is developed based on that data. There are two types of machine learning algorithms.

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- Supervised Learning
- Unsupervised Learning

In supervised learning algorithm, the dataset used for training the algorithm is labelled. An algorithm is used to learn a mapping function that maps input variables to the output variables. The aim of this learning process is to approximate the mapping function in such a way that for a given input data, the mapping function should predict the output data. It is important to note that the incorrect labelling in the data set can negatively impact the effectiveness of the machine learning model. Some commonly used supervised machine learning algorithms include Artificial Neural Networks, Support Vector Machines, Logistic Regression, Naive Bayes, and random forest. In unsupervised learning, the dataset is not labelled, hence difficult to interpret. Therefore, the algorithm uses the training data to generate a structure by looking at relations in training data itself. Some of the commonly used unsupervised learning algorithms include k-means clustering, k nearest neighbors, and auto-encoders. Supervised machine learning algorithms like a Support Vector Machine (SVM) have been widely used to predict the soil moisture behaviour.

The prediction in the soil moisture was traditionally driven by technical indicators. Technical indicators are the mathematical formulas which, when applied to historical data give a sense of how the stock is going to behave in future. With the emergence of technologies like machine learning the prediction of soil moisture indices has improved significantly. The machine learning algorithms consider the past prices of the soil moisture and train the model for future predictions of soil moisture. A huge number of cases have been done to predict the soil moisture behaviour using machine learning. In the article, two machine learning algorithms, namely, SVM and Artificial Neural Network (ANN) has been considered to predict the soil moisture behaviour since these algorithms have been quite successful.

2 RESEARCH AND METHODOLOGY

This work proposes that soil moisture prediction technique with improved accuracy than the techniques used in the existing literature. The proposed techniques use different parameters (thermogravimetric method) for the prediction of the soil moisture. Different machine learning classifiers are used.

2.1 The IoT Device made for measuring the soil moisture, temperature and humidity of the surrounding

The data collected in this work is by using a self-made IoT device consisting of ESP8266 MCU with Hygrometer Sensor, Temperature and Humidity sensor attached to it, please check Figure 1. The proposed techniques use different parameters (thermogravimetric method) for the prediction of the soil moisture.

Hardware Connections - When power is applied to the module you should see the red power light turn on and the blue serial indicator light flicker briefly. If you have a 3.3V FTDI Serial to USB board you can get started without fear of destroying your new ESP8266 WiFi module. Do note that many FTDI boards have a solder jumper to convert from 3.3V to 5V operation so ensure it is set to enable 3.3V operation.

ESP8266 SDK and Custom Firmware - An official Software Development Kit (SDK) has been released for the System-on-Chip (SoC) controller which powers the ESP8266 WiFi module. Using the SDK it's possible to add extra features to the AT command firmware or even create a standalone firmware.

NODE MCU based Smart Wi Fi Soil moisture indicator

Soil Moisture Sensor - Soil moisture is basically the content of water present in soil. This can be measured using a soil moisture sensor which consists of two conducting probes that act as a probe. It can measure the moisture content in the soil based on the change in resistance between the two conducting plates. The resistance between the two conducting plates varies in an inverse manner with the amount of moisture present in the soil.

Measuring soil moisture in terms of percentage. The analog output of soil moisture sensor is processed using ADC. The moisture content in terms of percentage is displayed on the serial monitor. The output of the soil moisture sensor changes in the range of ADC

Table 1: Collection of Data

No.	District	Place	Crop	Latit-ude	Long-itude
1	North 24 PGNS	Khardah	Mus-tard	22.9262901°	88.0015687°
2	Nadia	Chakdah	Paddy	22.9670786°	88.4156977°
3	Hoogly	Baseberia	Paddy	23.0079503°	88.4084241°
4	Howrah	Kalara	Paddy	22.5651529°	88.1189186°
5	Purba Mednipur	Kolaghat	Paddy	22.4370117°	87.8453062°
6	Paschim Mednipur	Daspur I	Potato	22.5274459°	87.7456503°
7	Bankura	Kotulpur	Potato	22.9174327°	87.6228784°
8	Purba Bardhaman	Raina II	Mustard	22.9732272°	87.7774093°
9	South 24 PGNS	Narendrapur	Cauliflower	22.4448646°	88.3947829°
10	Murshidabad	Bharatpur II	Potato	23.7268365°	88.0612147°
11	Birbhum	Chitgram	Mustard	23.7331991°	87.8658701°
12	Paschim Bardhaman	Piariganj	Potato	23.5402219°	87.4673144°
13	Kolkata	Rabindranagar	Cauliflower	22.653053°	88.403008°

value from 0 to 1023. This can be represented as moisture value in terms of percentage using formula given below.

$$AnalogOutput = \frac{ADCValue}{1023}$$

Moisture in percentage = 100 − (Analog output * 100)

For zero moisture, we get maximum value of 10-bit ADC, i.e. 1023. This, in turn, gives 0% moisture.

The probable advantage of using this IoT Device for the Farmers:

Water is an essential component for the development of plants in agriculture or irrigation. The paper stresses on the need of an externally hosted cloud computing platform to manage the database, android and the isolated server by the users across the country for irrigation. The system proposed in this paper uses information and communication technologies, allowing the user to consider and examine the information obtained by different sensors. Here we are using different sensors like humidity, temperature, moisture, light etc. These sensors give signal to the micro controller. Micro-controller gives the data to the isolated server through a serial communication. According to sensor values will be displayed on PC and Smart phone side. In this we keep threshold value for each sensor. The data is sent and processed on an isolated server, which stores the information from the sensors in a database, allowing further interpretation of data in a simple and flexible way. The intended system may lead to enhance the farming practices, overcoming the water crises and developing an upgraded agricultural system for the country.

2.2 Data collection and representation

Different machine learning classifiers are used. The data consists of 4 kinds of Kharif crops – Potato, Mustard, Paddy and Cauliflower, collected across 13 Districts of West Bengal, please check following Table 1

The data is a collection of about 90 days from 5th January 2020 up to 31st March 2020. The raw data only contains the Temperature, Humidity and Moisture, which are to be used in this research. Hence programs are to be written to calculate the other technical indicators in order to maximize the accuracy of the prediction of stock market trend. The machine learning algorithms that were used are Naïve Bayes, SVM, linear regression, and PCA. The machine learning models were created and evaluated using two different techniques. In the first one uses train- test split strategy. The training-testing split is simple the dataset is divided into two parts, i.e., training part and testing part. 25%, 50% and 75% of the data is used for both training and testing. It provides an efficient way for the machine learning algorithms. We need to find F1 Score is used to measure train and test accuracies of the algorithms.

2.3 Experimental setup and workflow

The programming language that was used in this work is python. The machine learning algorithms that were used are Naïve Bayes, SVM, linear regression, and PCA. The machine learning models were created and evaluated using two different techniques. In the first one uses train- test split strategy and the second one uses validation curve. In validation of the curve, this is the proper way of choosing multiple hyperparameters of an estimator on course grid search or similar methods (as Tuning the hyper-parameters of an estimator) which selects the hyperparameter with the maximum Score on a validation set or multiple validation sets. The hyperparameters were based on a validation score, whereas the validation score is biased and not good for estimation for the generalization any longer. In order to get a proper estimation for the generalization, we have to compute the Score on another test set. The training-testing split is simple the dataset is divided into two parts, i.e., training part and testing part. 25%, 50% and 75% of the data is used for both training and testing. It provides an efficient way for the machine learning algorithm. Kindly check the Figure 2, for the workflow of the experimental setup.

2.4 Algorithms used

Application of machine learning algorithms to determine the behaviour of the soil moisture is being deployed recently. This work used Naïve Bayes, SVM, Linear Regression, and PCA to predict the stock market trends. Naive Bayes is a classification algorithm with binary (two-class), and multi-class classification problems, the

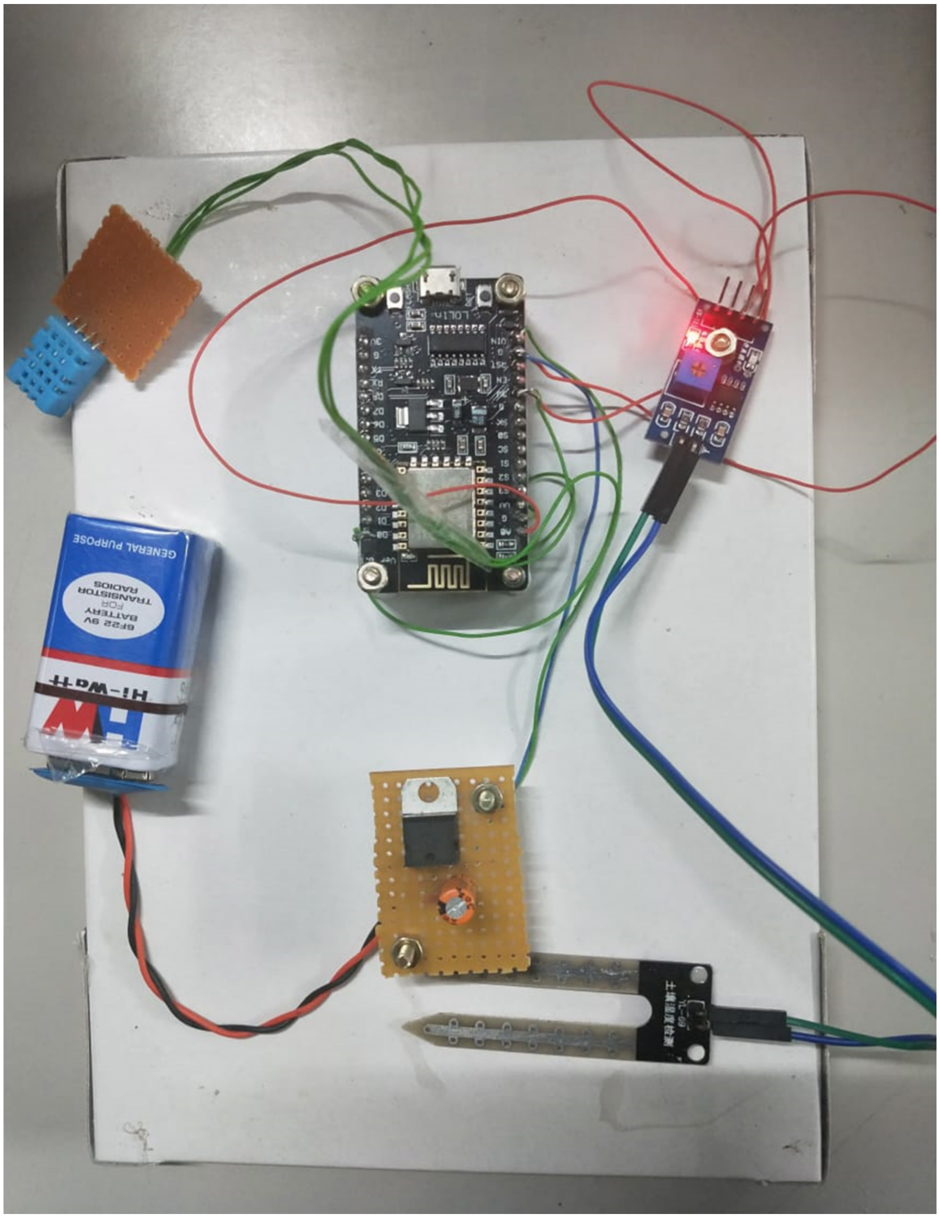

Figure 1: The IoT Device made is used to study Hygrometer, Temperature and Moisture

calculation of the probabilities are for each hypothesis those are simplified to make their calculation tractable. Linear Regression is a machine learning algorithm based on supervised learning, on performing regression task, regression models are used for predicting the outcome of an event based on the relationship between variables obtained from the dataset. Principal component analysis (PCA) is a technique to bring out strong patterns from a dataset by suppressing variations. It has been used to clean data sets to make it easy to explore and analyze. Support Vector Machines (SVM) are supervised algorithms used in machine learning. They are used in classification problems. Support Vector Machines also have a regression version called Support Vector Regression (SVR). It classifies the data by demonstrating linear separability in higher dimensions by using hyperplanes. When used for classification of data, SVM constructs an optimal separating hyperplane in the high dimensional feature space. In this way, it works like a maximal marginal classifier. When used for regression SVM classifier performs linear regression in the high dimensional vector space.

Finding - F1 Score - It is used to measure a test's accuracy. It is the Harmonic Mean between precision and recall. The range is from 0 to 1. It tells us how precise our classifier is that is how many instances it classifies correctly, and how robust it is that is, it does not miss a significant number of instances. High precision but lower recall, gives an extremely accurate score, but it then misses a

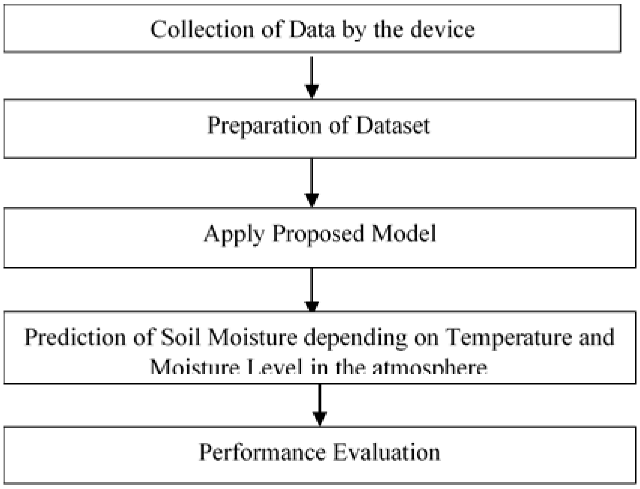

Figure 2: Workflow of the Experimental Setup

large number of instances which are difficult to classify. The greater the F1 Score, the better is the performance of the model. F1 Score tries in finding the balance between the precision and recall.

3 RESULTS AND DISCUSSION

3.1 Results based on Naïve Bayes Theorem

The first experiment was conducted to implement the Naïve Bayes on the dataset. The Naïve Bayes was implemented using 25%, 50% and 75% train – test split method on every crop's dataset. The performance parameters are calculated by taking the mean of the ten test splits. SVM was also implemented by dividing the data into training and test splits. The first type of implementation is done on the data after conversion into test-size supervised learning format. The result perceived from this experiment is that the accuracy of GNB classifier on training set is 0.09 and the accuracy of GNB classifier on test set is 0.04.

3.2 Results based on Linear Regression

The second algorithm considered in this work for prediction of soil moisture is linear regression. The linear regression, classification algorithm is also implemented by dividing the dataset in the training - testing splits of n_permutation upto 25, 50 and 75 on every crop's dataset. Here again, the machine learning model is implemented after conversion of the data into time series supervised learning format. The result perceived from this experiment is stated as the accuracy of Linear Regressor on training set is 0.38 and the accuracy of Linear Regressor on test set is 0.19

3.3 Results based on Support Vector Machine

The third experiment conducted was to implement the Support Vector Machine (SVM) on the dataset. The SVM was implemented using cross-validation method crop's dataset. The performance parameters are calculated by taking the mean of the ten test splits. SVM was also implemented by dividing the data into training and test splits. The third type of implementation is done on the data after conversion into time series supervised learning format. The result perceived from this experiment is the accuracy of SVM classifier on both the training set and the test set is 0.08. In non-time series supervised learning data, comparing cross-validation method with the train-test method of SVM implementation, it can be is observed that the performance parameters do not vary with any degree of significance. This also indicates the fact that issues like overfitting and under-fitting are not a cause of concern for the data sets that are being used in this work.

3.4 Results based on PCA

The last method used to predict the soil moisture is Principal Component Analysis, is to reduce the dimensionality of any data set consisting of many variables which are correlated with each other, either heavily or lightly, or while retaining the variations present in the dataset, till the maximum extent. Similarly, is done by transforming the variables into a new set of variables, which are known as the principal components and are orthogonal, ordered such that the retention of variation present in the original variables decreases as moved down in the order. Logistic regression considers how each independent variable impact on response variables. The result perceived from this experiment is that the accuracy of PCA classifier on training set is 0.09 and that of accuracy of PCA classifier on test set is 0.04

The accuracies that are obtained show that SVM is the best model, than Linear Regression models perform better than PCA and Naïve Bayes models. This indicates that Linear Regression has three layers of Neural Networks, thus performing very well with linear kernel. The PCA and Naïve Bayes models perm better as both of them are two layered Neural Network and is in binary classification. Naïve Bayes does not entirely handle the complexities of the soil moisture data since linear regression is a single layer neural network. The PCA also performed well than Naïve Bayes model as it is a dimension reduction of the problem while retaining as much as information possible. The SVM model performs very well with linear kernel since the problem is the binary classification problem. Naïve Bayes does not entirely handle the complexities of the soil moisture data since linear regression is a single-layered Neural Network. Also, once the algorithms are applied to the data in the aspect of soil moisture supervised learning format, the accuracies have been increased, and errors have been decreased. Thus, it will help the farmers to adjust their management strategy beforehand.

4 CITING RELATED WORK

Having studied various papers related to soil moisture prediction and analysis. The discussion of some research articles are provided below:

(Vyas. et al. 1995)[4] This paper has total experimentation done in arid and semi-arid region of Rajasthan, India, non-irrigated Indian Mustard is often grown during winter on the conserved moisture of low lying areas. The different levels of stored soil moisture of 100%, 90%, 75%, 60% of field capacity has been observed with decreasing moisture levels. The parameter used and the functions used are as Mean, LSD, moisture content of the soil for the production of plants raised under different levels of stored moisture.

(Cai. et al. 2019)[5] They have projected a deep learning regression network with data fitting capability, which was proposed to construct the soil moisture prediction model by integrating the dataset, analyzing the time series of the predictive variables. By the result, we get to know that deep learning model is feasible and effective for soil moisture prediction. It's good data fitting, and generalization capability will enrich the input characteristics while ensuring high accuracy in predicting the trends and value of soil moisture data and proves theoretically for water-saving irrigation and drought control. For the verification of the superior performance of the DNNR model, it is compared to the existing models of machine learning algorithms, where the soil moisture prediction models mostly used are SVM(Support Vector Machine), LR(Linear Regression), ANN(Artificial Neural Network) and DNNR. The DNNR model constructed here is superior to the above model in the evaluation of the comparison with multiple performance measures. In a non-stationary time series, which presents a periodic variation with regular pattern involving large fluctuations in soil moisture prediction. From correlation analysis, it is known that each parameter characters have a correlation with the moisture parameter, which affects the predicted value, and the initial soil moisture features have the greatest weight is the primary. Humidity and temperature are secondary. The deep learning model used is to predict the soil moisture at a depth of 20 cm in an area. It was then proven by experimentation that too many layers of the model could lead to too excessive training time and overfitting, whereas overfitting can affect training accuracy and generality. Finally, the two-layer hidden layer was considered most suitable for their model's structure. The first layer is always responsible for learning the input features, and the second layer follows on responsibly for polynomial fitting of the learned features, and too many nodes would cause overfitting and reduced the prediction accuracy and generalization capability. At the same time, the DNNR model could also predict the moisture trends of the other two regions, and which has the ability to keep prediction error almost near to the zero point. All evaluations indicates that the MLP model is the worst of all the models. The above results indicates that the DNNR model has excellent generalization capability and scalability. It is feasible enough to apply soil moisture prediction and provide technical support for irrigation strategies and drought control using this model.

(Prakash., et al., 2019)[10] There has been three different datasets used for the prediction of soil moisture, upon using Multiple Linear Regression, SVM and Recurrent Neural Network. This paper has introduced machine learning techniques for prediction of soil moisture in advance, and concluded that multiple linear regression is superior to SVM and RNN.

The previous papers stated that Simple Binary Classification Problem, SVM, Non-Linear Regression, and basic use of SVM through MATLAB had been used by the author. Ensembling Floating Adaptive Basis Function Construction, Bagging and Boosting Predictions have shown some success in predicting the model successfully. The use of deep learning method that performs better than traditional statistical and physical models for reliable estimation of SM based on remotely sensed satellite data. To build an effective deep learning model, we examined the influences of sampling criteria and input parameters as well as the accuracy of several deep neural networks. The selected model was cross-validated to determine its stability. To build an effective deep learning model, we examined the influences of sampling criteria and input parameters as well as the accuracy of several deep neural networks. The selected model was cross-validated to determine its stability. There has been three different datasets used for the prediction of soil moisture, upon using Multiple Linear Regression, SVM and Recurrent Neural Network. This paper has introduced machine learning techniques for prediction of soil moisture in advance, and concluded that multiple linear regression is superior to SVM and RNN.

Another work by (Lee. et al., 2018)[18] states that they have used Weka forecasting library to forecast soil moisture based on SVM Regression for prediction framework to estimate and forecast soil moisture level over time based on meteorological data is parsed from the ICN database and fed to machine learning algorithms for training and validating purposes. The sparse and well-studied machine learning techniques SVM regression applied to the historical data to derive mathematical models. Designed from a Precision Agriculture perspective, the sites specific model is able to incorporate data from other sources at the granularity of one day. The feature of taking user-provided data makes the system more robust by allowing the model to interact with human knowledge or reliable soil moisture data from other sources at a fine granularity. The machine learning algorithms requires a large data size, field-collected data from our sensor node are not used in our prediction experiments. However, the soil and meteorological attributes collected from the hardware devices are the same attributes that are used for deriving prediction models.

5 CITING RELATED WORK

This section cites a variety of journal [2, 4, 19], conference [3, 14, 17, 22, 23], and magazine [8–12, 17] articles to illustrate how they appear in the references section. It also cites books [1, 6, 13, 15, 16, 21], a technical report [11], a workshop presentation [18], an online reference [7], an article [5] and a PhD dissertation [20].

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